

Warmup: Cross validation

Model building and cross validation

Your friend collects training data of size $n = 500$ observations, on 1000 different explanatory variables.

They have too many explanatory variables available, so they use forward stepwise selection (with AIC) to choose a model, limiting the number of variables allowed in the model to 25.

In the end, they have a regression model with 25 explanatory variables. Let's call this their `selected_model`. To assess the predictive ability of their `selected_model`, they use cross validation in the following way:

- Divide the their training data into 10 folds
- Fit the `selected_model` (with the same explanatory variables as those chosen by stepwise selection) to 9 of the 10 folds, and predict on the remaining fold
- Repeat, and average across the 10 folds

Here is their code, and the resulting estimate of prediction error:

```
# start with only an intercept
initial_fit <- glm(y ~ 1, data = train_data, family = binomial)

# add predictors one at a time based on AIC
selected_model <- step(initial_fit,
  scope = formula(glm(y ~ .,
    data = train_data,
    family = binomial)),
  direction = "forward", trace = 0,
  steps = 25)

# 10-fold cross validation estimate of prediction error
k <- 10
boot::cv.glm(train_data, selected_model,
  cost = logistic_deviance, # loss function to use
  K = k)$delta[1] * k/nrow(train_data) # average deviance
```

```
[1] 1.09569
```

Later, they collect a separate test sample, independent of the training sample, which is from the same population. Here is the average deviance calculated on this new test data:

```
# predicted probabilities on test data
pred_probs <- predict(selected_model, newdata = test_data,
                      type = "response")

# estimate of test error using test data
logistic_deviance(test_data$y, pred_probs)/n_test
```

```
[1] 2.131921
```

Question: Their 10-fold CV estimate of prediction error is *not* representative of the true prediction error on new data. Why? What should they do differently?

Hint: The code is a faithful implementation of their proposed process. The problem is with the process itself.